network address translation (Nat)

المحاضرة السادسة

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Network address Translation(NAT)

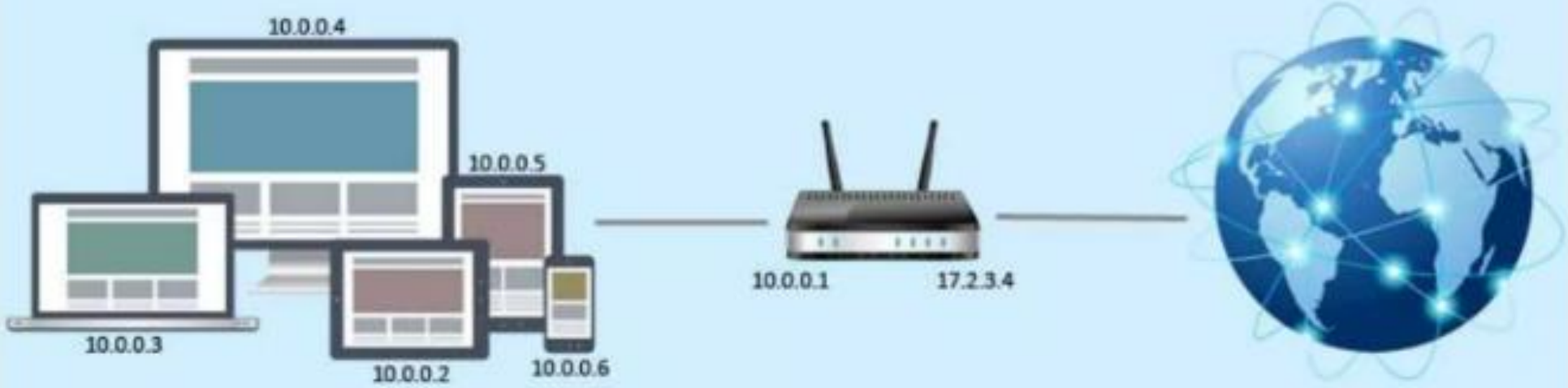
OUTLINE

- ✓ what is Nat
- ✓ why we use Nat
- ✓ working of Nat
- ✓ Types of Nat
- ✓ Advantage/dis of Nat

Network Address Translation(NAT)

NAT is a technique the that is use to translates the private IP addresses into public IP addresses AND public IP addresses to private IP addresses.

Network Address Translation



Why we use NAT

- Whatever connects directly into internet must have public unique IP addresses .
- IPV4 contains around 2.4 billion IP addresses.
- There is a shortage of IPV4 addresses

The solutions

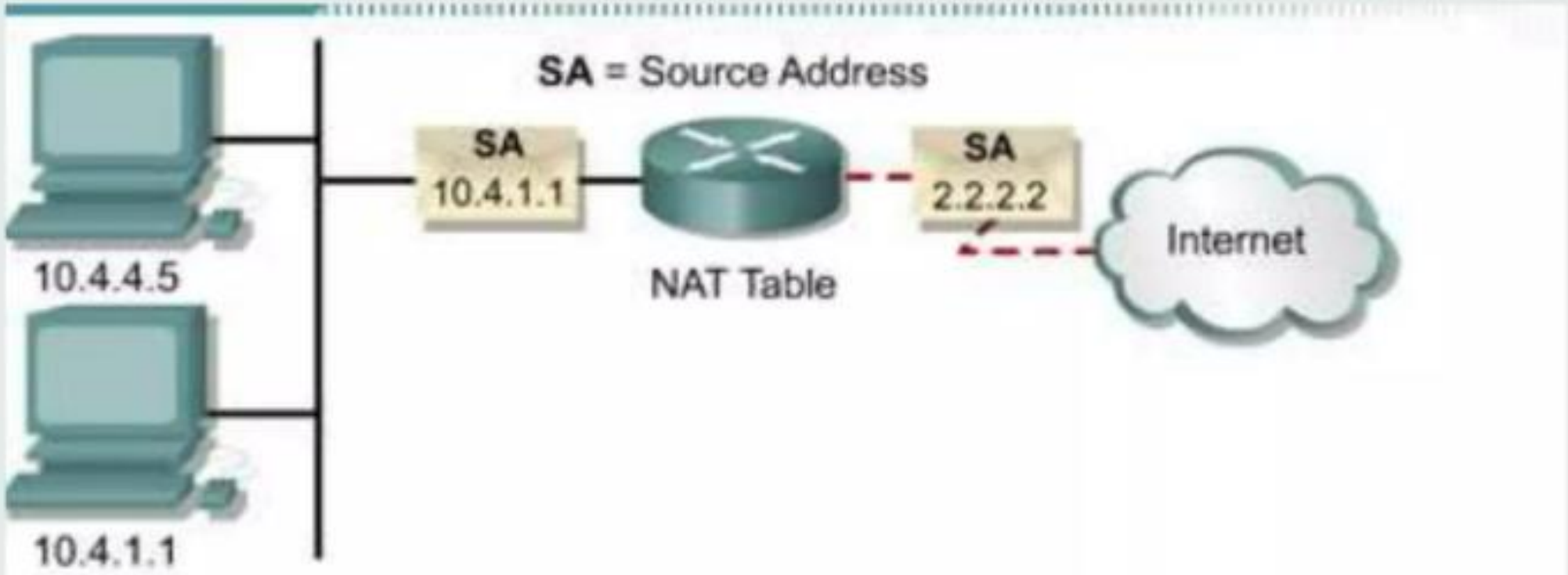
- Long term solution: IPV6
- Short term solution is CIDR(classless inter domain routing)
- NAT is another solution

Why we use NAT

- Network Address Translation allows a single device, such as a router, to act as an agent between the Internet (or "public network") and a local (or "private") network.
- This means that only a single, unique IP address is required to represent an entire group of Computers.

So the shortage of IP addresses is only one reason to use NAT.

Working of NAT



Inside Local IP Address	Global IP Address
10.4.4.5 TCP Source Port 1026	~~2.2.2.3~~ 2.2.2.2 TCP Source Port 1923
10.4.1.1 TCP Source Port 1026	~~2.2.2.2~~ 2.2.2.2 TCP Source Port 1924

Types Of NAT

1. Static NAT
2. Dynamic NAT
3. Port Address Translation(PAT)

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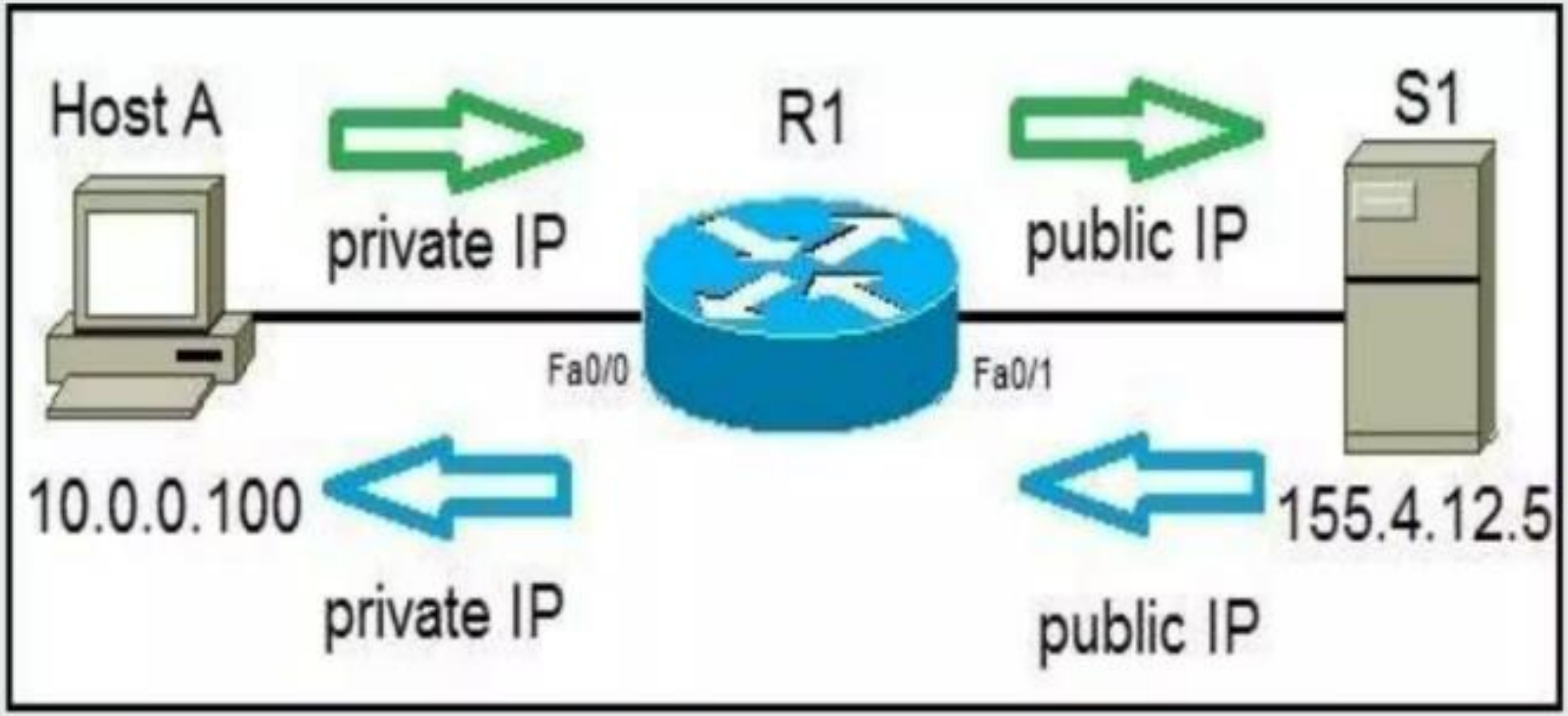
Static NAT

Static NAT perform one-to-one translation between two address ,or between a port on one address to a port on other address. Static NAT is most often use to assign a public address to a device behind a NAT-enable firewall/router.

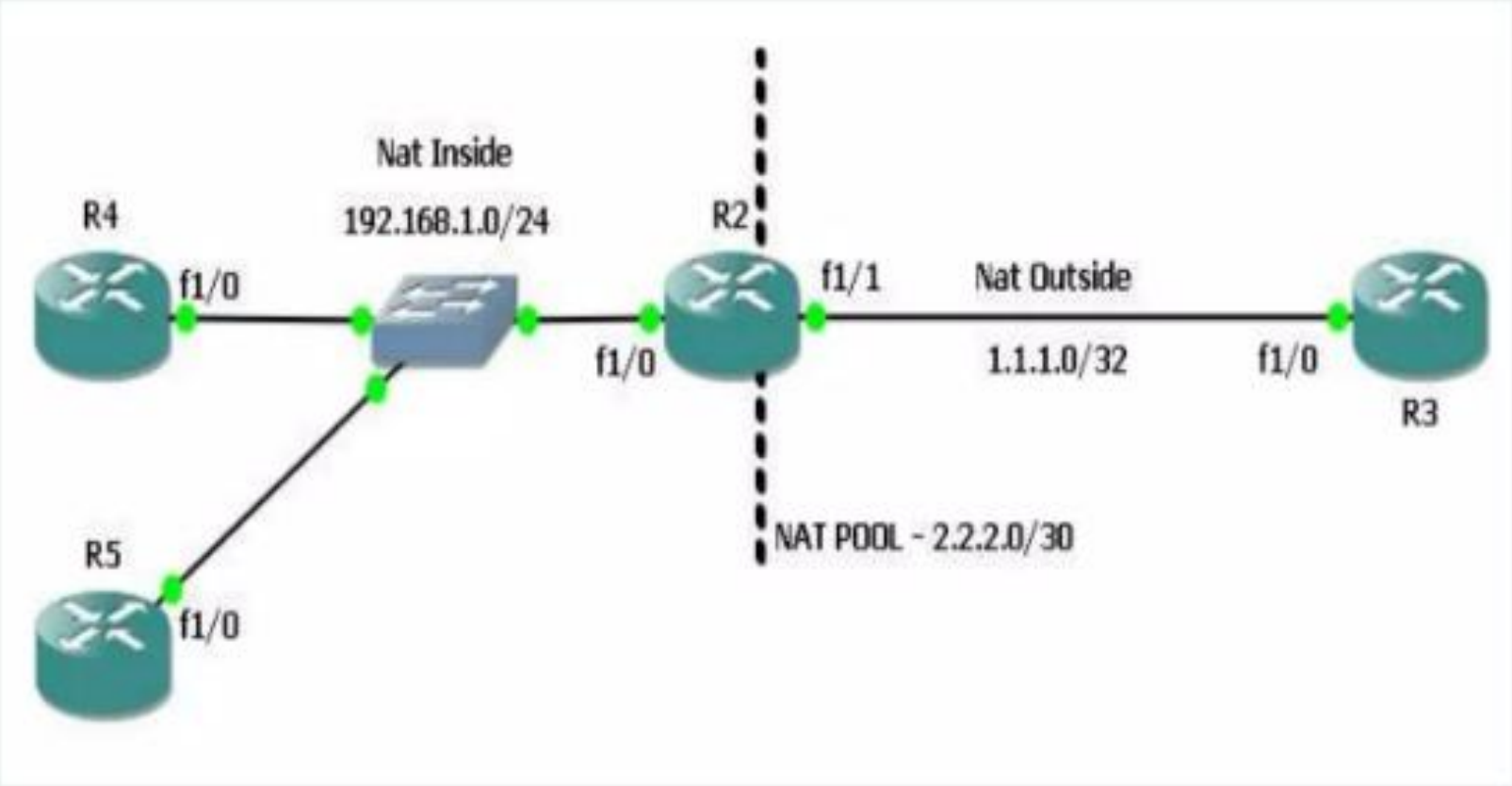
Static NAT


Host A
R1
S1
private IP
public IP
Fa0/0
Fa0/1
10.0.0.100
155.4.12.5
private IP
public IP

Dynamic NAT

Dynamic NAT utilizes a pool of global addresses to dynamically translate the outbound traffic clients behind a NAT-enabled device.

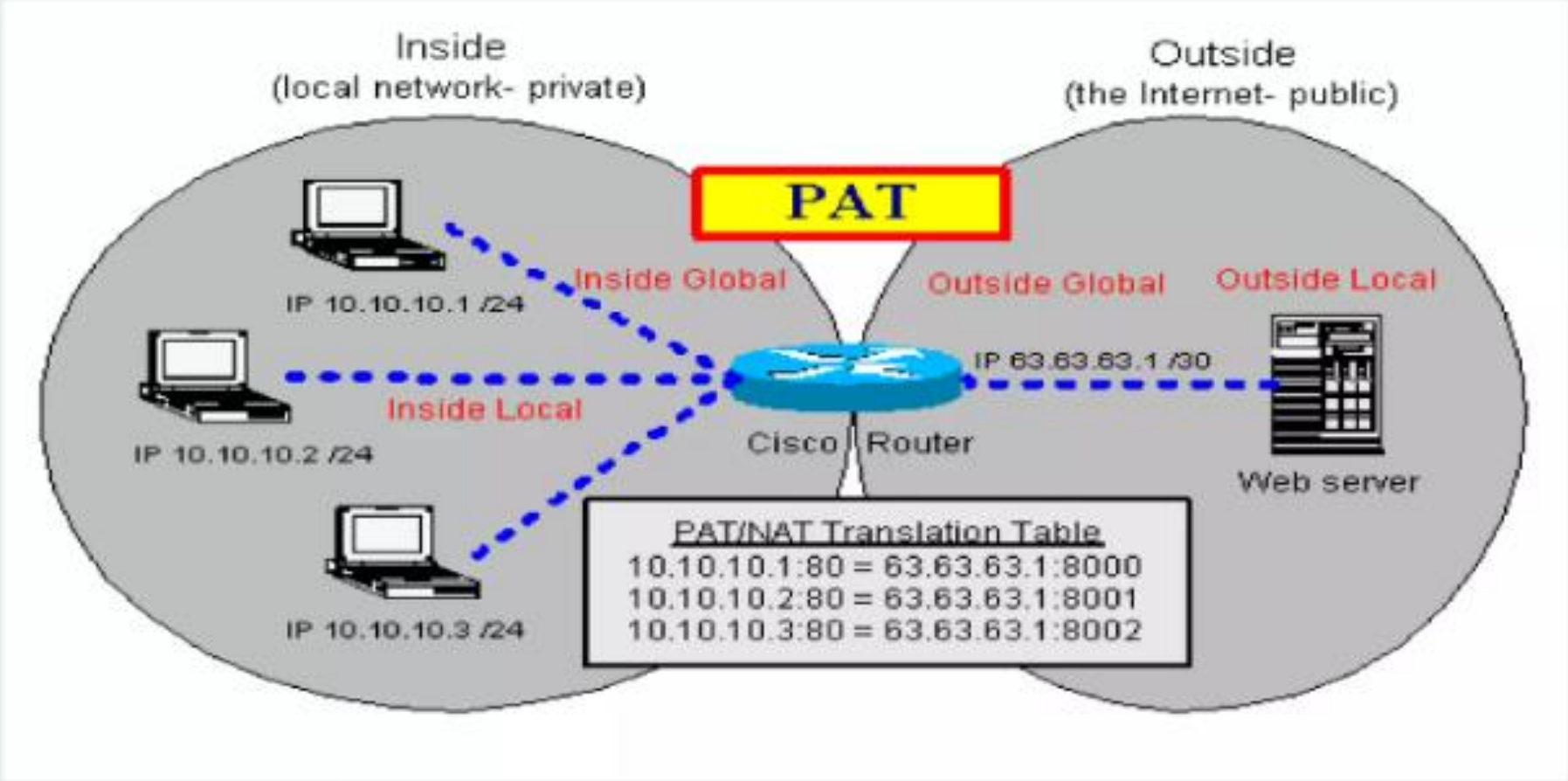
Dynamic NAT


Nat Inside
192.168.1.0/24
R4
f1/0
R2
f1/1
Nat Outside
1.1.1.0/32
f1/0
R3
f1/0
R5
f1/0
NAT POOL - 2.2.2.0/30

Port Address Translation(PAT)

NAT overload or port address translation(PAT) translate the outbound traffic of clients to unique port number off of a single global address. PAT is necessary when the number of internal clients exceeds the available global addresses.

Port Address Translation(PAT)



Advantages Of NAT

- It hide the real IP address of your internal network from the public network and act as a firewall.
- It allows unlimited number of private addresss to access the single internet connection.
- Hence , it allows multiple devices to access single internet connection.It helps you to save money from buyimg multiple internet connection for multiple devices.

Disadvantages Of NAT

- If you change the IP address, the troubleshooting may become more complex.
- It blocks some incoming connection.
- Some TCP/IP applications like peer to peer application ,end to end IPsec, multicast routing protocol do not work well with NAT.